

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0689

Ninemile 100 kV
September 15, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0689. The Interconnection Customer proposes a 176.7 MW natural gas combined cycle facility interconnecting at the Ninemile 100 kV substation in the Ohio area. The request was received on October 14, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 337 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,648,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
COLLINWOOD to NINEMILE 100 kV	NINEMILE to WEYERHAUSER SNOQUALMIE MILL 100 kV L	162.0	95	100.0
NINEMILE SUBSTATION to WEYERHAUSER SNOQUALMIE MILL 100 kV	NINEMILE to COLLINWOOD 100 kV Line	126.9	121	100.0
MORRISTOWN to WINGER 100 kV Circuit 1	WINGER to MORRISTOWN 100 kV Circuit 2	105.1	150	6.7
MORRISTOWN SUBSTATION to WINGER 100 kV Line 2	MORRISTOWN to WINGER 100 kV Cir. 1	105.1	150	6.7
20 PANT SUBSTATION to WINGER 100 kV Line	WALTERIA 500 kV/230 kV Transformer Bank	102.5	132	7.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
COLLINWOOD to NINEMILE 100 kV	Rebuild line with higher-capacity conductor	\$1,268,000	100% (\$1,268,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WYERHAUSER SNOQUALMIE MILL to NINEMILE SUBSTATION 100 kV	Rebuild line with higher-capacity conductor	\$1,646,000	100% (\$1,646,000)
MORRISTOWN to WINGER 100 kV Line 1	Reconductor line	\$340,000	100% (\$340,000)
WINGER to MORRISTOWN SUBSTATION 100 kV	Reconductor line	\$283,000	100% (\$283,000)
20 PLANT to WINGER 100 kV	Reconductor line	\$354,000	31% (\$111,000)
	Total allocated to project		\$3,648,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.